

SOLAR ENERGY



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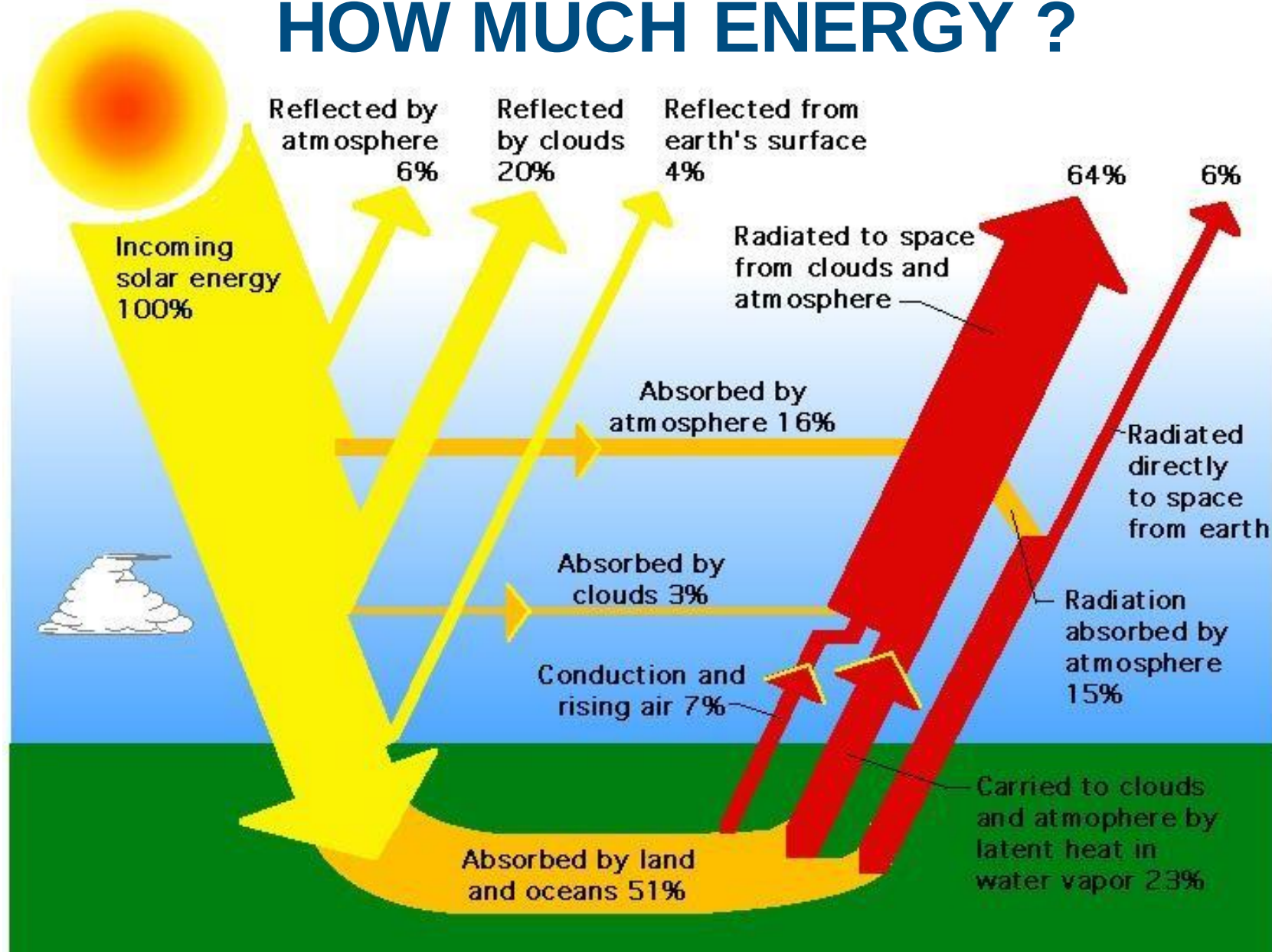
INTRODUCTION :

There are two types of energy sources :

- 1.Non-Renewable sources of energy :- which are available in limited quantity. (Conventional sources of energy)
Eg:- Natural gas, Coal, Oil, etc.
- 2.Renewable sources of energy :- which can be renewed naturally over time.
Eg:- Solar energy, Wind energy, Tidal energy, etc.

SOLAR ENERGY is radiant light and heat from the Sun that is harnessed using a range of technologies such as solar power to generate electricity, solar thermal energy.

HOW MUCH ENERGY ?



WHAT IS SOLAR POWER ?

Solar power is the conversion of renewable energy from sunlight into electricity, either directly using photovoltaics (PV) or indirectly using concentrated solar power, or a combination. Concentrated solar power systems use lenses or mirrors and solar tracking systems to focus a large area of sunlight into a small beam.

SOLAR POWER SYSTEM COMPONENTS

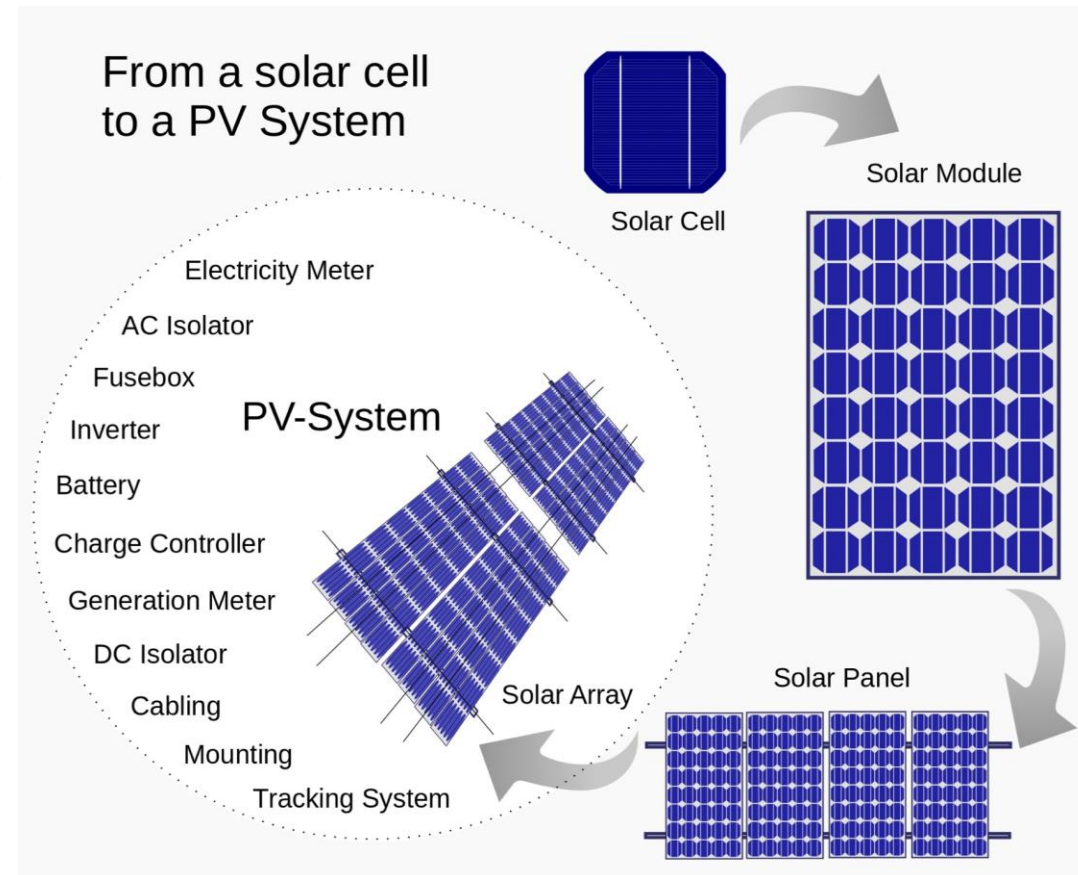
- SOLAR PANELS
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SOLAR PANELS

A solar panel or a photo-voltaic module is an assembly of photo-voltaic cells mounted in a framework for installation. Solar panels use sunlight as a source of energy to generate direct current electricity. A collection of PV modules is called a PV panel, and a system of PV panels is called an array. Arrays of a photovoltaic system supply solar electricity to electrical equipment.

A solar cell is configured as a large area p-n junction made from silicon.

The illuminated side of a solar cell generally has a transparent conducting film for allowing light to enter into the active material and to collect the generated charge carriers. Typically, films with high transmittance and high electrical conductance such as indium tin oxide, conducting polymers or conducting nanowire networks are used for the purpose.



BATTERIES

- A battery system is essential to compensate for the absence of any electrical output from the cells at night or in overcast conditions.
- Each battery stores direct current (DC) electricity at a fixed voltage determined by the panel specifications.
- Eg:- Lithium Manganese Oxide (LMO)
Lithium Nickel Manganese Cobalt Oxide (NMC)
Lithium Nickel Cobalt Aluminum Oxide (NCA)
Lithium Iron Phosphate (LFP)



CHARGE CONTROLLER

- In high sun exposure the voltage produced by the solar panels has the potential to damage to batteries.
- A charge controller is required to regulate the charge; ultimately increasing the battery life and performance.



SOLAR INVERTER

- Solar cells generate DC current.
- A solar inverter or PV inverter, is a type of electrical converter which converts the variable direct current (DC) output of a photovoltaic (PV) solar panel into a utility frequency alternating current (AC) that can be fed into a commercial electrical grid or used by a local, off-grid electrical network.

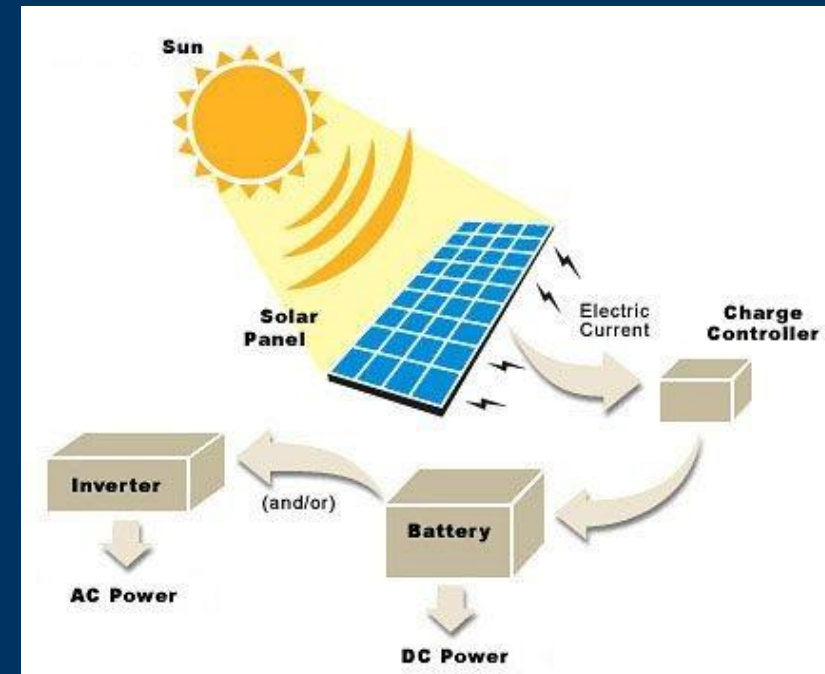
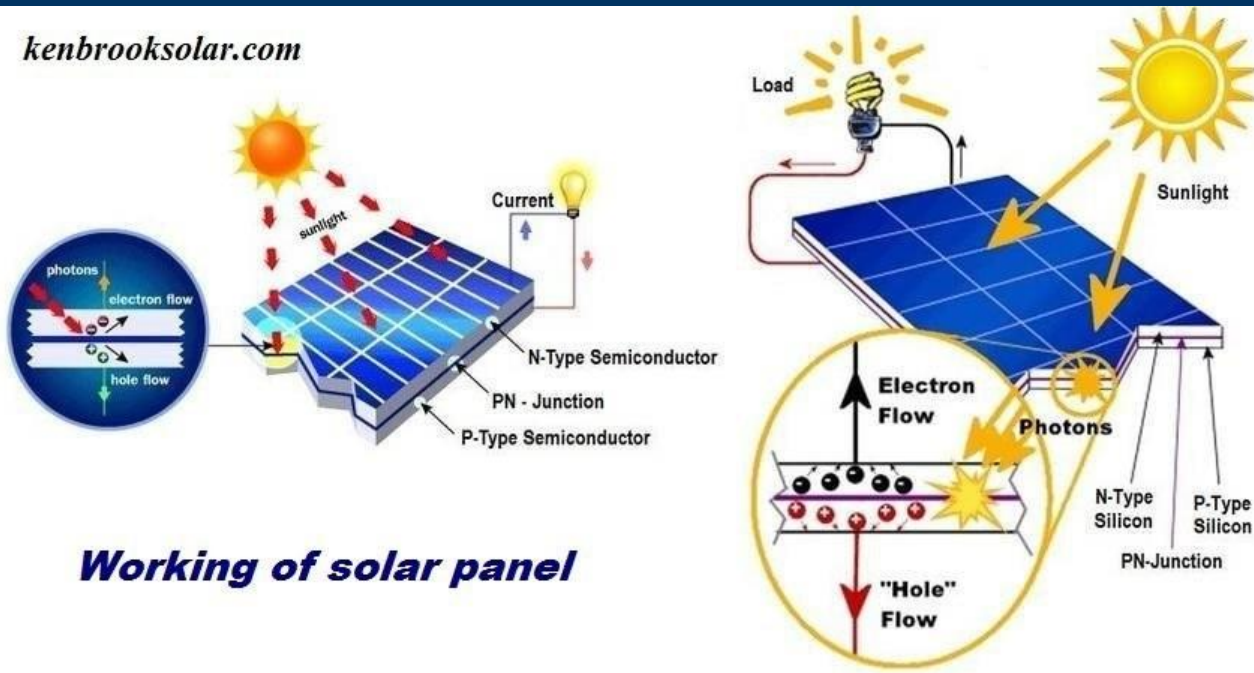


WORKING OF PHOTOVOLTAIC SYSTEM

Photovoltaic system works on the principle of photovoltaic effect.

When light falls on a solar cell, electrons in the absorber layer are excited from a lower-energy “ground state,” in which they are bound to specific atoms in the solid, to a higher “excited state,” in which they can move through the solid. In the absence of the junction-forming layers, these “free” electrons are in random motion, and so there can be no oriented direct current. The addition of junction-forming layers, however, induces a built-in electric field that produces the photovoltaic effect. In effect, the electric field gives a collective motion to the electrons that flow past the electrical contact layers into an external circuit where they can do useful work.

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ADVANTAGES OF SOLAR POWER

- Solar energy is a clean and renewable energy source.
- Once a solar panel is installed, solar energy can be produced free of charge.
- Solar energy will last forever
- Solar energy causes no pollution.
- Solar cells make absolutely no noise at all.
- Very little maintenance is needed to keep solar cells running.
- There are no moving parts in a solar cell which makes it impossible to really damage them.
- In the long term, there can be a high return on investment due to the amount of free energy a solar panel can produce, it is estimated that the average household will see 60% of their energy coming in from solar panels.

DISADVANTAGES OF SOLAR POWER

- Solar panels can be expensive to install resulting in a time-lag of many years for savings on energy bills to match initial investments.
- Electricity generation depends entirely on a country's exposure to sunlight; this could be limited by a country's climate.
- Solar power stations do not match the power output of similar sized conventional power stations; they can also be very expensive to build.
- Solar power is used to charge batteries so that solar powered devices can be used at night. The batteries can often be large and heavy, taking up space and needing to be replaced from time to time.
- Efficiency of pv system is less than 20%. ie, less than 20% of the solar energy absorbed by the pv panel gets converted into electrical energy.